Banting Lecture 1995

A Lesson in Metabolic Regulation Inspired by the Glucokinase Glucose Sensor Paradigm

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Special features of glucose metabolism in pancreatic β-cells are central to an understanding of the physiological role of these cells in glucose homeostasis. Several of these characteristics are emphasized: a high-capacity system for glucose transport; glucose phosphorylation by the high-K_m glucokinase (GK), which is rate-limiting for glucose metabolism and determines physiologically the glucose dependency curves of many processes in β-cell intermediary and energy metabolism and of insulin release and is therefore viewed as glucose sensor; remarkably low activity of lactate dehydrogenase and the presence of effective hydrogen shuttles to allow virtually quantitative oxidation of glycolytic NADH; the near absence of glycogen and fatty acid synthesis and of gluconeogenesis, such that intermediary metabolism is primarily catabolic; a crucial role of mitochondrial processes, including the citric acid cycle, electron transport, and oxidative phosphorylation with F_oF_1 ATPase governing the glucose-dependent increase of the ATP mass-action ratio; a Ca^{2+}-independent glucose-induced respiratory burst and increased ATP production in β-cells as striking manifestations of crucial mitochondrial reactions; control of the membrane potential by the mass-action ratio of ATP and voltage-dependent Ca^{2+} influx as signal for insulin release; accumulation of malonyl-CoA, acyl-CoA, and diacylglycerol as essential or auxiliary metabolic coupling factors; and amplification of the adenine nucleotide, lipid-related, and Ca^{2+} signals to recruit many auxiliary processes to maximize insulin biosynthesis and release. The biochemical design also suggests certain candidate diabetes genes related to fuel metabolism: low-activity and low-stability GK mutants that explain in part the maturity-onset diabetes of the young (MODY) phenotype in humans and mitochondrial DNA mutations of F_oF_1 ATPase components thought to cause late-onset diabetes in BHE_{cdb} rats. These two examples are chosen to illustrate that metabolic reactions with high control strength participating in β-cell energy metabolism and generating coupling factors and intracellular signals are steps with great susceptibility to genetic, environmental, and pharmacological influences. Glucose metabolism of β-cells also controls, in addition to insulin secretion and insulin biosynthesis, an adaptive response to excessive fuel loads and may increase the β-cell mass by hypertrophy, hyperplasia, and neogenesis. It is probable that this adaptive response is compromised in diabetes because of the GK or ATPase mutants that are highlighted here. A comprehensive knowledge of β-cell intermediary and energy metabolism is therefore the foundation for understanding the role of these cells in fuel homeostasis and in the pathogenesis of the most prevalent metabolic disease, diabetes. *Diabetes* 45:223–241, 1996

In the preface to the first issue of the journal *Diabetes*, Eliot P. Joslin cited two quotes that are worth remembering and are also well suited as prescript for a Banting Lecture (1): "Hitch your wagon to a star!" and "Disce ut semper victurus, vive ut cras moriturus."

The first refers to Sir Frederick Banting, whose handsome portrait graces the cover of the inaugural issue of *Diabetes* and who remains an inspiring role model for many diabetes researchers. The saying could also relate to the research topic a scientist chooses for a life's task. Glucokinase (GK) has played such a role in my scientific career, and I am glad to have followed its brilliant light for more than a quarter of a century. The second saying is about learning and living. I experience the award of the Banting Medal as a powerful impetus to continue learning everything possible about the pancreatic β-cell, and the purpose of this lecture is to include others in this exciting quest. The quote is, of course, also a salutary reminder of one person's limitations.

The main purpose of our work is to understand the role of the pancreatic β-cell in glucose homeostasis and in the pathogenesis of diabetes. In this endeavor, it is imperative to pay close attention to the other factors that are vital in regulating blood glucose. This includes foremost the insulin receptor system and the counterregulatory neuroendocrine system. The cyclical nature of the feedback loops that sustain glucose homeostasis and the features of reciprocity and complementarity of β-cell functions when contrasted with the characteristics of counterregulatory endocrine cells and target tissues of insulin provide many important clues as to how the β-cells might be designed biochemically. One striking manifestation of this coordinated reciprocal relationship is the dose-response curves of glucose stimulation and glucose inhibition of β- and α-cells, respectively. The opposite dose-response curves do cross over exactly at the

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DAG, diacylglycerol; DG, deoxyglucose; G-6-P, glucose-6-phosphate; G-6-Pase, glucose-6-phosphatase; GK, glucokinase; GLP-I, glucagon-like peptide I; IBMX, isobutylmethylxanthine; MODY, maturity-onset diabetes of the young; NMR, nuclear magnetic resonance; P_i, inorganic phosphate; PCR, polymerase chain reaction; RT, reverse transcriptase.

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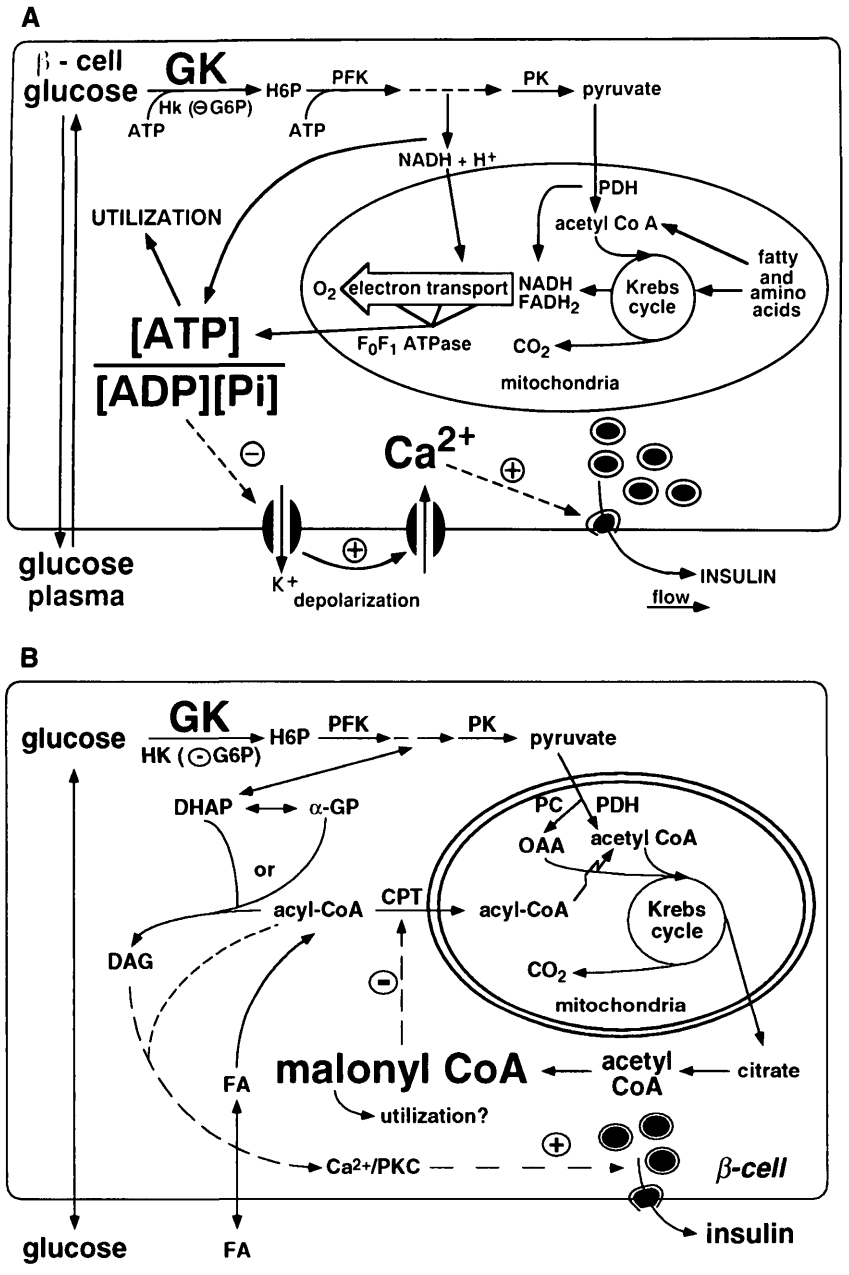


FIG. 1. Crucial steps in glucose-induced insulin secretion. *A:* generation of adenine nucleotide signals. *B:* generation of lipid-related signals.

euglycemic level of 5 mmol/l glucose. An expression of complementarity is the enhanced removal of glucose, amino acid, and fatty acid from the blood due to insulin, when release of the hormone is stimulated by these same nutrients. Extrapolations from the clinical arena indicate drastic distortions of these physiological relationships in diabetes. It is the objective of our research to discover the biochemical basis of such fundamental physiological and pathological phenomena.

The present discussion focuses on the role that intermediary and energy metabolism of the β-cell plays in glucose stimulation of insulin release, and it highlights the implications of our basic knowledge in this area for diabetes. GK is central to our discussion of this topic because this enzyme explains essential features of glucose-stimulated insulin release in humans and common laboratory animals (2–5). The enzyme's singular role in glucose stimulation of insulin release is captured befittingly by the term *glucokinase glucose sensor*. High-capacity glucose transport and GK of islet tissue were first demonstrated directly by Matschinsky and Ellerman (6) in 1968. The investigations of Grodsky et al. (7) and Coore and Randle (8) had already shown that metabolism of glucose was required for stimulation of secretion to occur. The presence of GK in human pancreatic islet tissue was established by Bedoya et al. (9), allowing the formulation of a general hypothesis that a deficiency of this enzyme in β-cells could cause diabetes in humans. The GK genes of rats and humans were cloned through the efforts of several laboratories in the late 1980s and early 1990s (10–15), and linkage between the GK gene and the maturity-onset diabetes of the young (MODY) form of NIDDM was uncovered in 1992 by Froguel and colleagues (16,17) and by other groups (18). About 30 GK mutations have since been detected and characterized by several laboratories (5,19,20). GK is considered as important a diabetes gene as insulin and the insulin receptor, yet together they explain only a very small fraction of this disease (probably <5%). Their exploration has, however, contributed tremendously to our understanding of normal and impaired metabolic regulation.

The lessons learned from studying the role of GK as β-cell

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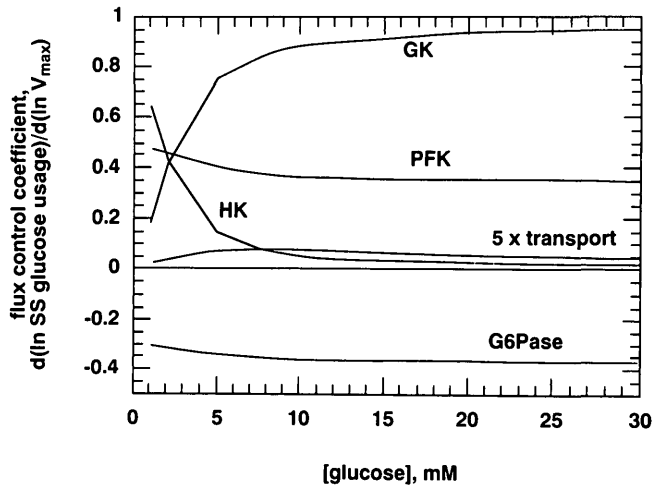


FIG. 2. Sensitivity of glucose usage to maximal activities of glucose transport, GK, hexokinase, G-6-Pase, and P-fructokinase using reasonable assumptions and published data about islet cell glucose metabolism. The sum of all control coefficients is 1. A value of 1 indicates that a step has the highest control strength, qualifying it as a pacemaker step. From Sweet and Matschinsky (44).

glucose sensor and as candidate gene for NIDDM provide useful directives in our search for other candidate diabetes genes encoding enzymes of intermediary and energy metabolism of β-cells. The genes of the F_oF_1-ATPase complex, which catalyses an essential step in metabolic coupling in β-cells, are suggested as novel candidate genes for NIDDM and are therefore chosen here as second example to illustrate the strategy.

A MARVELOUS MOLECULAR MACHINE

To understand glucose homeostasis and its pathology in diabetes, it is necessary to develop a clear picture of the biochemical design of the insulin-producing pancreatic β-cells. It is now clear that the unique features of glucose metabolism must be central to any realistic biochemical portrait of this cell (2–5,21,22). A brief description of the β-cell as a molecular machine emphasizing its glucose and energy metabolism is therefore the point of departure for this presentation (Fig. 1*A*). In engineering terms, the β-cell is viewed as a glucostat in analogy to a thermostat and GK, the pacemaker of glucose metabolism, is conceptualized as the glucose sensor in analogy to a temperature-sensitive element in a thermostat. Practically all other considerations in this overview emanate from or are related to this cardinal tenet of β-cell physiological chemistry. In an illuminating comparable case, the carotid body cytochrome a_3 has been identified as the oxygen sensor (23).

Metabolism of glucose is conclusively required for stimulation of insulin release and biosynthesis. Metabolism and stimulation of insulin secretion by various hexoses is characterized by well-defined substrate specificity, typical concentration dependencies, and preference of the α-anomers, which can be explained readily by the kinetic characteristics of GK. Other design features of glucose metabolism fit the concept that GK is the pacemaker of several sequences of reactions leading from the substrate glucose to many products, including pyruvate, lactate, oxaloacetate, citrate, malonyl-CoA, CO_2, H_2O, and ATP. These processes translate changes of blood glucose levels into the rate of intracellular signal formation primarily by mass action. To begin with, glucose is transported into the β-cell by high-capacity low-affinity Na^+-independent carriers. GLUT2 is the predominant carrier in rodent β-cells, but it remains to be seen whether this is universally true, in particular whether GLUT2 is essential for β-cell function in humans. The glucose carrier system, whatever its molecular nature, is efficient enough to assure rapid equalization between extra- and intracellular glucose, such that hyperglycemia and hypoglycemia are precisely mirrored by the intracellular glucose level. This feature by itself suggests an intracellular glucose sensor site rather than a membrane-associated receptor as is apparently true for the taste of sweetness. In addition to GK, β-cells also seem to contain hexokinase, probably hexokinase I. The capacities in β-cells of these two enzymes appear to differ by an order of magnitude, with GK playing a predominant role. Their affinities for glucose differ by 3 orders of magnitude. The K_m of GK is reported to range from 6 to 11 mmol/l, and the substrate concentration dependency is sigmoidal, with a Hill number of 1.4–1.8, which manifests itself in an inflection point of 2.5–4.0 mmol/l. In β-cells, GK is apparently not controlled by direct or indirect product inhibition. The cosubstrate of the phosphorylation reaction is Mg^+-ATP, which is present in β-cells at near-saturation levels. These features of GK together with the fact that glucose transport capacity exceeds it by about 100-fold are consistent with its pacemaker role and are the hallmarks of an ideal glucose sensor system. The fate of glucose-6-phosphate (G-6-P) is governed by the characteristic enzyme complement of the β-cell. Glycolysis predominates, and the pentose-P shunt is quantitatively of minor significance (i.e., <10% of flux). Normal β-cells appear to lack classical glucose-6-phosphatase (G-6-Pase) (24,25); they also do not contain a prominent enzymatic apparatus for glycogen metabolism. P-fructokinase is the next potential regulatory step, as is universally true for glycolysis. The formidable array of known positive and negative regulators seems to be of little consequence in the control of glycolysis in normal β-cells because hexokinase and G-6-Pase are quantitatively insignificant and because GK is not inhibitable by its product. Fructose-2,6-P_2 appears to be quantitatively unimportant as P-fructokinase activator. Fructose-6-phosphate as substrate and fructose-1,6-P_2 as product of the enzyme are powerful allosteric activators of P-fructokinase, overriding any feedback inhibition by ATP and other intermediates. The glycolytic intermediate pool from fructose-1,6-P_2 to P-enolpyruvate is in equilibrium, and pyruvate is generated at a rate determined by fructose-1,6-P_2, P-enolpyruvate, and ADP levels. Its rate of production keeps pace with glucose phosphorylation. β-cells produce hardly any lactate because lactate dehydrogenase activity is very low (26) and because cytosolic NADH levels are probably also low because of efficient shuttles that transfer hydrogen into mitochondria (21). Cytosolic hydrogen seems to be channeled nearly quantitatively into the electron transport chain through the α-glycero-P and malate-aspartate shuttles. Electron transport from NADH or reduced flavin adenine dinucleotide ($FADH_2$) to cytochrome c appears to be in equilibrium, and the level of cytochrome c in turn determines the actual rate of cytochrome *c* oxidase (27). Electron transport is coupled to proton pumping across the mitochondrial membrane, and the proton motive force thus generated (ΔpH and $\Delta\psi$) results in ATP synthesis (28,29). Complex V, or the F_oF_1 ATPase, thus plays a central role in metabolic coupling. From the well-docu-

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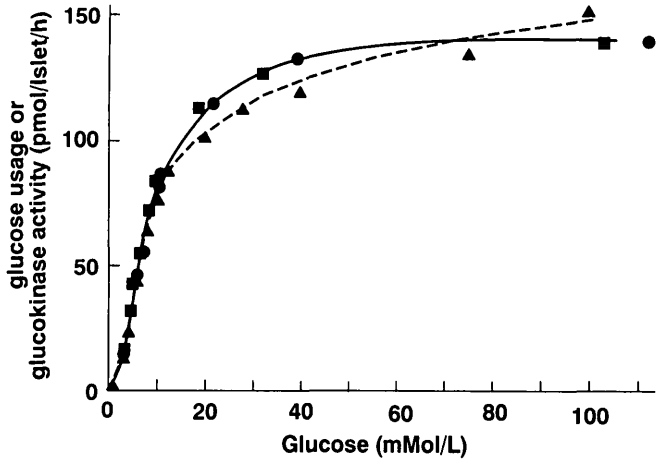


FIG. 3. GK determines the rate of glycolysis. The curves relating the rate of glucose phosphorylation due to GK in the islet homogenate and the rate of the high-K_m component glucose usage measured with intact islets as a function of the glucose concentration are virtually indistinguishable. From Meglasson and Matschinsky (3).

mented glucose-induced enhancement of respiration (30), it can be calculated that the metabolism of each glucose molecule increases ATP generation by 6–8 molecules (including glycolytic ATP and depending on the relative contribution of the shuttles). The ATP mass-action ratio (ATP/ADP × P_i [inorganic phosphate]) is increased by glucose metabolism and controls the open-state probability of adenine nucleotide–sensitive potassium channels, which results in slow membrane depolarization (31,32). The crucial signal for this effect appears to be the level of free ADP (33,34). A high mass-action ratio of ATP may also drive the calcium ATPase of the endoplasmic reticulum, explaining a clearly demonstrable transient fall of the cytosolic Ca^{2+} when β-cells are suddenly exposed to high glucose (35,36). When a critical threshold potential is reached, action potentials develop that depolarize the cell membrane and open voltage-sensitive Ca^{2+} channels (31). Ca^{2+} influx is enhanced manifold, and cytosolic Ca^{2+} levels are elevated (37). The rise of intracellular Ca^{2+} due to increased influx may be further augmented by voltage-dependent release from intracellular Ca^{2+} stores (38).

A crucial point to be emphasized here is that the respiratory burst, the enhanced ATP generation, and the initial slow membrane depolarization that are all caused by exposure to high glucose are independent of extracellular Ca^{2+} and do not seem to require cytosolic Ca^{2+} to rise (36). This important fact frequently is not understood or is ignored. Enhanced glucose oxidation is, however, partly dependent on high extracellular Ca^{2+}, suggesting differential control of electron transport and ATP synthesis on one hand and pyruvate dehydrogenase and the citric acid cycle on the other (36).

It has been proposed that certain metabolic coupling factors related to lipid metabolism may play an essential or auxiliary role in glucose-induced insulin release in addition to adenine nucleotides (39–43). High glucose elevates malonyl-CoA, long-chain acyl-CoA, and diacylglycerol (DAG) in β-cells (Fig. 1*B*). Acyl-CoA and DAG may activate in a Ca^{2+}-dependent manner certain protein kinases that participate in exocytosis and thus enhance insulin secretion. In the context of the present discussion, it is hypothesized that physiologically these effects of glucose are also governed by

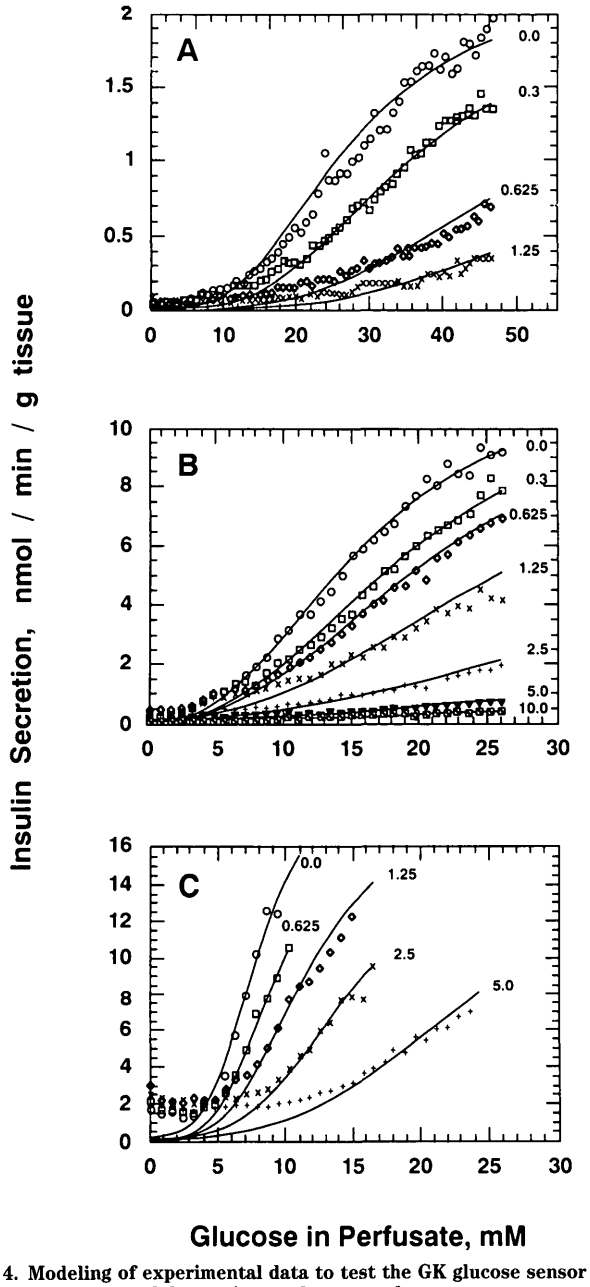


FIG. 4. Modeling of experimental data to test the GK glucose sensor concept. Experimental data points and computed curves are presented. Each data point is the mean of at least three determinations. The continuous curves represent the modeled data. Graded inhibition of GK by mannoheptulose causes predictable inhibition of insulin release in isolated perifused islets cultured at 4, 10, and 30 mmol/l for 7 days.

the GK glucose sensor and that increased substrate pressure through mass action is the primary driving force for generating them, assisted by allosteric mechanisms. Oxaloacetate (generated by pyruvate carboxylation and as a product of the malate/aspartate shuttle) condenses in the mitochondrial matrix with acetyl-CoA to form citrate, which may be transported into the cytosol to serve as precursor of acetyl-CoA. Acetyl-CoA is carboxylated to malonyl-CoA, a reaction allosterically activated by citrate, and malonyl-CoA is believed to switch off mitochondrial acyl-CoA usage, causing its accumulation in the cytosol. Acyl-CoA directly by itself and/or indirectly by serving as building block of DAG is thought to serve as metabolic signal. It is not known whether the production of these lipid-related coupling factors requires intracellular Ca^{2+} to rise or whether it is an independent

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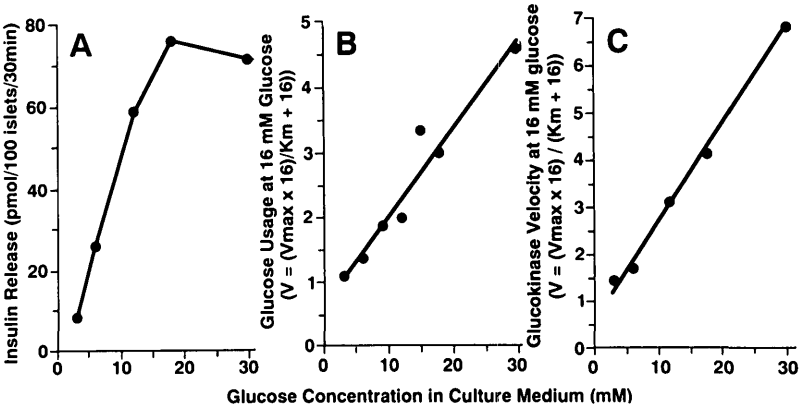


FIG. 5. Concordant alterations of GK, glucose usage, and insulin release in isolated islets cultured at different glucose concentrations. From Liang et al. (45).

primary event comparable to the glucose-induced change of the adenine nucleotide trigger(s). It still remains to be established whether the proposed involvement of these lipid-related coupling factors is essential for glucose-induced insulin release.

Other fuel stimulants of the β-cells, for instance a physiological mixture of 20 amino acids or long-chain fatty acids, require physiological concentrations of glucose to be effective stimuli for insulin release. This is also true for all stimulatory neurotransmitters and gut-derived hormones, such as acetylcholine and glucagon-like peptide I (GLP-I). Glucose metabolism and its coupling to secretion needs to be thoroughly understood because of this requirement in order to fully explain the action of other glucose-dependent fuel and nonfuel agonists.

With this minimal conceptual model of glucose and glucose-dependent agonist stimulation of insulin secretion as a basis, an in-depth discussion of a few selected aspects of the system promises to be more plausible. This minimal model also provides a framework for the discussion of the role that an abnormal β-cell might play in the pathophysiology of diabetes, in particular of NIDDM.

CYTOSOLIC STEPS OF GLUCOSE METABOLISM: GENERATION OF NADH, PYRUVATE, AND ATP

Cytosolic glucose metabolism in β-cells might be introduced instructively by a brief account of a sensitivity analysis of steady-state glucose usage to glucose transport, GK, hexokinase, P-fructokinase, and G-6-Pase using mathematical modeling (44). We have calculated the flux-control coefficients for these steps using reported data and reasonable assumptions about the rate coefficients of the corresponding enzymes (Fig. 2). It is apparent that GK exhibits a high control strength consistent with its role as pacemaker for glycolysis, even when net flux is reduced because of futile cycling. It is also obvious that transport influences flux very little. G-6-Pase is expected to result in a decrease of net glucose usage. Its coefficient is thus negative but is nearly mirrored by the positive coefficient for P-fructokinase. This analysis points to the early steps of glucose metabolism as pacesetting reactions and also suggests a regulatory role for P-fructokinase provided G-6-Pase is functional. Similar considerations about the potential role of P-fructokinase apply to the GK regulatory protein that has been claimed to be present in β-cells. In this presentation, we will focus our attention on GK and G-6-Pase.

There are many experiments in the literature that document that GK determines the rate of glucose usage and insulin release in islet tissue and in certain tumorous β-cell lines. One striking example is that of Meglasson and Matschinsky (3) showing that the glucose dependency curves of islet GK and of glucose use are virtually indistinguishable (Fig. 3). This close relationship between GK activity and glucose use is maintained under varying physiological circumstances, for instance in response to feeding-starvation regimens and in aging (2–5). Another illustration of the correlation stems from a recent meticulous reassessment of mannoheptulose inhibition of glucose use and insulin release (I.R. Sweet, G. Li, H. Najafi, F.M.M., unpublished observations) (Fig. 4). Glucose metabolism and insulin release are studied with a minimal two-compartment model of the pancreatic islet, and the model is tested by its fit with experimental data from an extensive investigation of the effects of mannoheptulose as competitive inhibitor of GK. The close fit between model and experiment strongly endorses the view that GK is the pacemaker and sensor of the system. Mathematical modeling indicates that inhibition of hexokinase is not to be implicated under these conditions. The mannoheptulose sensitivity greatly increases when islets are cultured in low glucose resembling a starvation condition, but mannoheptulose is less potent when islets have adapted to high glucose in culture. A disparity of modeled curves and the actual data at low glucose levels for islets that have been cultured in 30 mmol/l glucose is probably explained by a high basal rate of release, which might be due to persistent glucose potentiation, an aspect of β-cell physiology poorly understood and not included in the model. It is possible to induce the activity of β-cell GK manifold by culturing isolated islets in high glucose and to demonstrate that glucose use and insulin release are increased pari passu (Fig. 5) (45). Precise measurements of glucose phosphorylation in extracts of whole islets are greatly impeded by the fact that hexokinase may exceed GK by a factor of as much as fivefold. Most of the hexokinase is probably derived from non-β-cells and the high-K_m GK component may actually escape detection because of this contamination. This is particularly true for human islet tissue (9). The difficulty was overcome by a very sensitive radiometric oil-well method (46). In this procedure, hexokinase is inhibited by excess G-6-P and the reaction is carried out in submicroliter droplets under oil. The assay has a sensitivity of detecting as little as 1 pg of GK, sufficient to determine GK activity quantitatively in <100 β-cells. This assay helped in discovering that human pancreatic islet tissue has GK activities similar to those observed in that of mice and rats (9).

There is now wide agreement about GK measurements in

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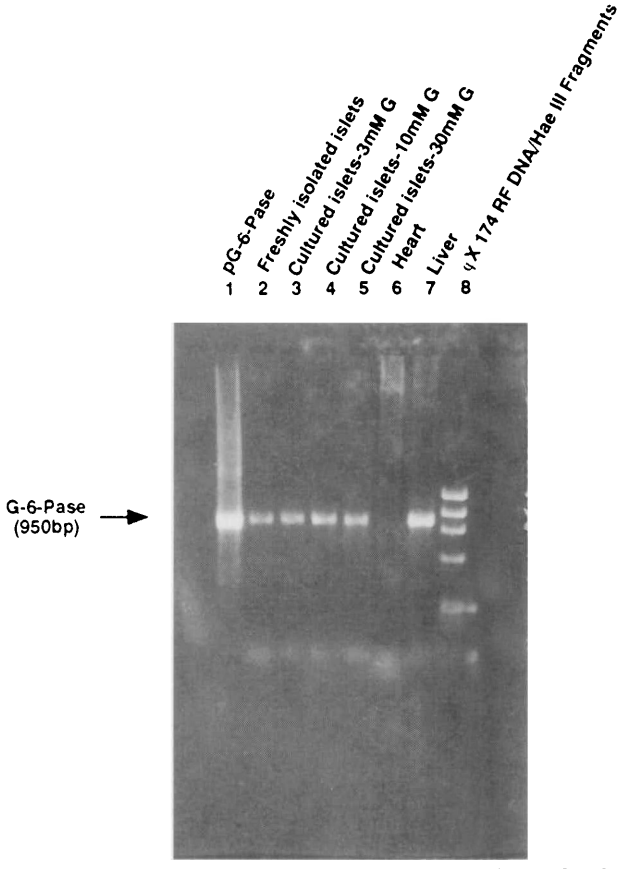


FIG. 6. Demonstration of mRNA for G-6-Pase in freshly isolated and cultured islet tissue and in control tissue by RT-PCR.

various islet preparations. It is reasonable to extrapolate from the results of these enzyme measurements to the actual glycolytic rate as it might exist in islets in situ at any given blood glucose concentration. This extrapolation is justified because GK is the pacemaker of glucose metabolism. One should, however, briefly pause here to reflect on the fact that we have absolutely no direct information about glucose metabolism of human pancreatic β-cells in situ. It is also noteworthy in this context that Lowry and colleagues (47) have developed quantitative histochemical techniques using 2-deoxyglucose (2-DG) to measure in situ glucose phosphorylation in microscopic structures of brain. It is probable that these techniques are applicable to islet tissue in situ, although difficulties of interpretation may arise from the cellular diversity of islet tissue and from the possible interference by a G-6-Pase and G-6-P dehydrogenase (see below).

Because of this close relationship between GK and glucose usage as determinants of glucose-induced insulin release, it seemed reasonable to conceptualize GK as glucose sensor and to predict that even small reductions of the enzyme's activity would result in a rightward shift of the threshold for glucose stimulation of insulin release and development of a diabetic phenotype (2–5). This prediction has been fulfilled by the genetic and clinical exploration of a certain subset of patients with MODY (see below).

Because of many similarities of glucose metabolism in β-cells and liver, it seemed reasonable to search for G-6-Pase in islet tissue. Many investigators found that various islet tissue preparations hydrolyzed G-6-P, yet uncertainties remained about the substrate specificity of this enzyme activity. The careful radiometric studies of Aktar Khan et al. (25) with intact islet tissue have inferred from the observation of

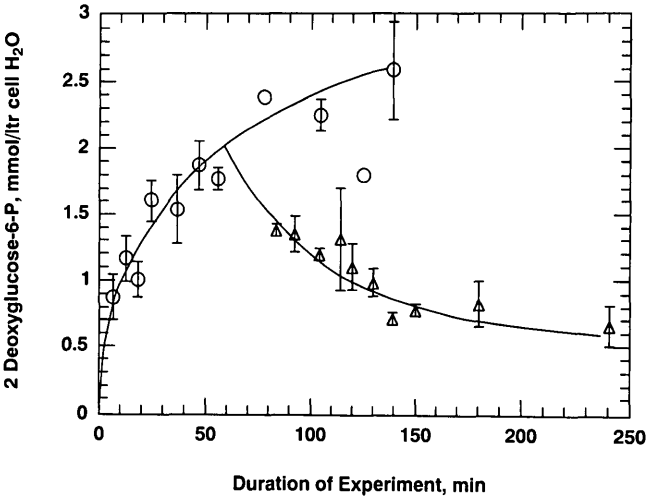


FIG. 7. Metabolism of 2-DG-6-P by intact freshly isolated rat islets. Islets were first incubated for 60 min in 30 mmol/l 2-DG and were then studied after switching to a medium that lacked the 2-DG but had 20 mmol/l mannoheptulose included to inhibit phosphorylation of residual DG. Controls remained in 2-DG. The disappearance of 2-DG-6-P was measured by enzymatic fluorometry in neutralized perchloric acid extract of islets.

futile cycling that a G-6-Pase–like enzyme could be operative in intact islets of diabetic mice but seemed not to be functional in islets from normal rodents. Malaisse and colleagues (24) have also found no evidence for G-6-Pase in cell-free preparations of normal rat islet tissue. Our results agree with the positions of these two laboratories (I.R. Sweet, P. Kesavan, R. Taub, H. Najafi, J. Grodberg, G. Li, F.M.M., unpublished observations). We found, however, a positive reverse transcriptase–polymerase chain reaction (RT-PCR) reaction for the mRNA of this enzyme in normal rat islets and various β-cell lines (Fig. 6). To further assess the physiological significance of this finding, we have attempted to detect G-6-Pase in unbroken normal islet cells. For this purpose, we have incubated intact isolated islets in high concentrations of 2-DG to generate large 2-DG-6-P pools

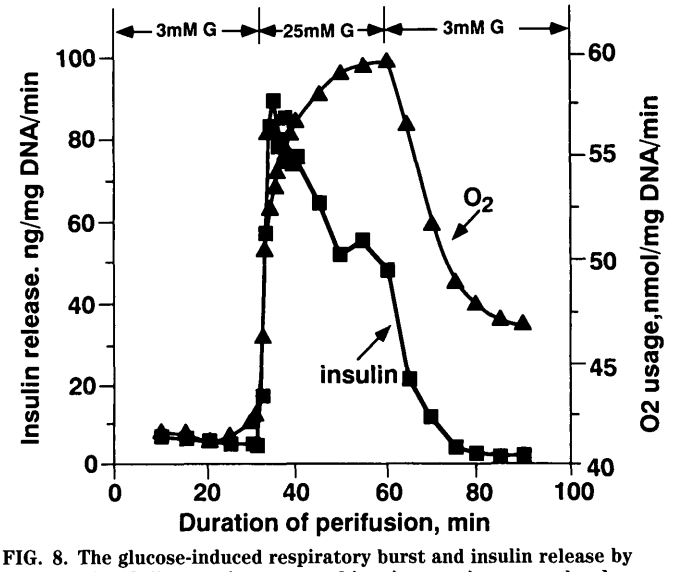


FIG. 8. The glucose-induced respiratory burst and insulin release by β-HC-9 cells. Cells were incorporated in microscopic agarose beads resembling pseudo islets and were perifused. The oxygen concentration was measured polarographically in the perifusate. DNA was used as a measure of cell number (10 μg DNA/10^6 cells). The glucose level was 3 mmol/l during the baseline and increased to 25 mmol/l during stimulation. IBMX (0.1 mmol/l) was included to enhance the secretory response. The data represent the means of four experiments.

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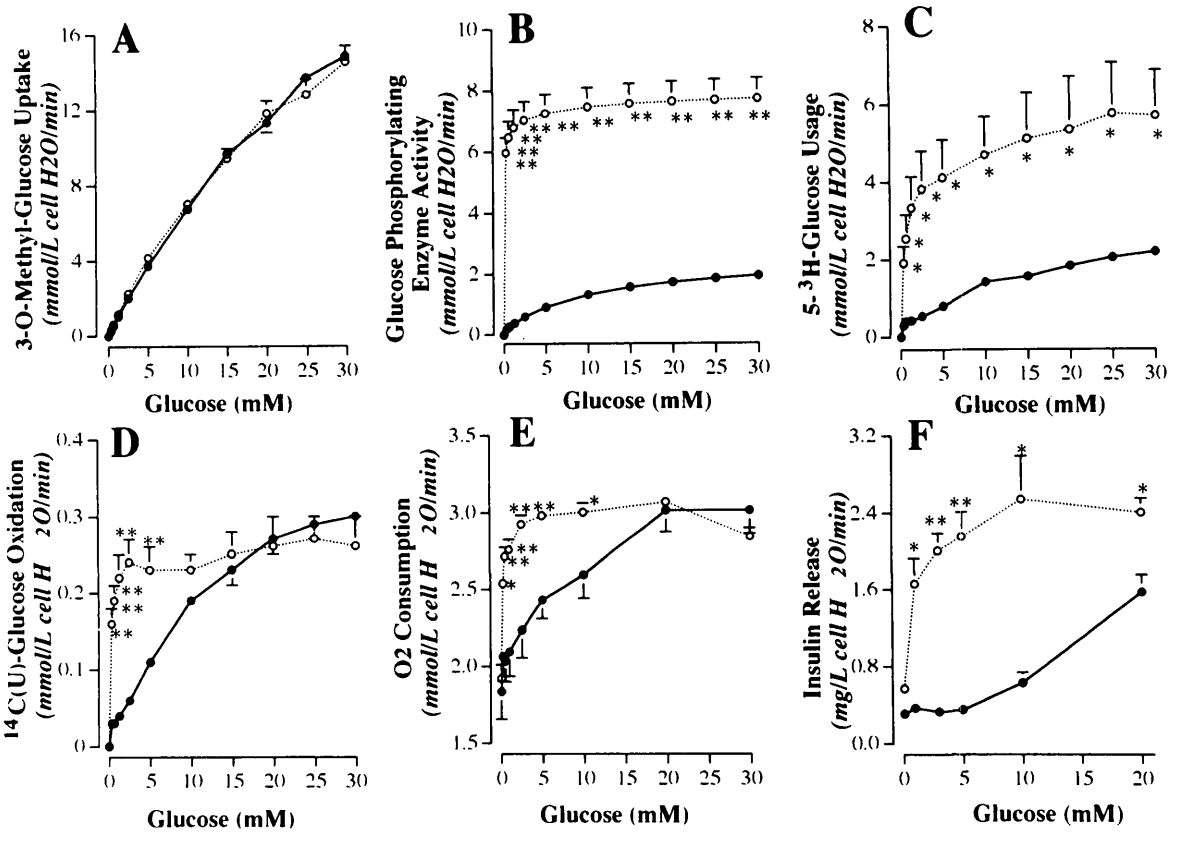


FIG. 9. Glucose concentration dependence of transport (*A*), phosphorylation (*B*), glucose usage (*C*), glucose oxidation (*D*), O_2 consumption (*E*), and insulin release (*F*) of β-HC-9 (solid curves) and β-TC-3 (dotted curves) cells. The means ± SE of at least three experiments are depicted.

as a model substrate for the putative phosphatase and have then followed the pools' depletion as potential evidence for the enzyme's activity (Fig. 7). It needs to be recognized, however, that this test is not totally specific because 2-DG-6-P may also be metabolized by other enzymes, for example by G-6-P dehydrogenase. We observed a halftime of 30 min or a *k* value of ~2% per minute for the process of 2-DG-6-P decay. The data suggest an enzymatic process with a K_m higher than 5 mmol/l, which could be explained by a G-6-Pase, glucose-6-dehydrogenase, or both. We found that a low-speed supernatant of isolated rat islets hydrolyzed G-6-P but that the enzyme or enzymes lack characteristics that are typical for hepatic G-6-Pase, such as latency and the ability of the intact microsomal fraction to distinguish between G-6-P and 2-DG-6-P. The enzyme is probably of little practical importance at physiological G-6-P levels of 25 to 50 μmol/l because the measured K_m for G-6-P and 2-DG-6-P is >10 mmol/l. It remains to be seen whether this approach will detect futile cycling of glucose as reported in animal models of diabetes. The present deliberations indicate that G-6-Pase is of minimal importance in normal islets, but it is likely that the enzyme may assume considerable significance in diabetes (25).

We need to pay less attention to the other glycolytic reactions between G-6-P to pyruvate because their control strength is inherently low or because their control strength is dependent on the presence of enzymes and elements that are not well established, e.g., the dependence of the control strength of P-fructokinase on G-6-Pase, the GK regulatory protein, or both (44). β-cells have little lactate dehydrogenase in contrast to most other tissues (26). Thus, pyruvate and NADH rather than lactate accumulate as chief end products of glycolysis. A detailed discussion of the fate of glycolytic NADH and of pyruvate is therefore the topic of the section to follow.

MITOCHONDRIAL STEPS: ROLE OF HYDROGEN SHUTTLES, THE CITRIC ACID CYCLE, ELECTRON TRANSPORT, AND OXIDATIVE PHOSPHORYLATION

The participation of mitochondrial reactions is crucial for glucose stimulation of insulin secretion, yet few direct studies have been performed with mitochondria from β-cells because of the scarcity of material. Much of our knowledge in this area is therefore indirect and much of it is speculative. It may be illuminating to introduce the topic with the description of the classical experiment demonstrating a respiratory burst of islets when exposed to high glucose, an experiment executed for the first time in the 1960s by Claes Hellerstrom with the Cartesian diver (48). Shown is a recent version of such an experiment (Fig. 8), which was performed with about 500 mg of cultured β-HC-9 cells (i.e., ~0.5 billion cells) immobilized in microscopic agarose beads that are used as pseudo islets. These artificial islets contain a homogeneous cell population, which greatly facilitates data interpretation, in contrast with the difficulties of evaluating studies of samples weighing much less than a milligram and containing cellularly diverse islet tissue. The β-cells were perifused for 30 min with 3 mmol/l glucose, stimulated with 25 mmol/l glucose for 30 min, and finally observed for another 30 min in low glucose. Oxygen consumption and insulin release were determined. Oxygen consumption was stimulated by ~50%, and insulin release increased more than 10-fold. The stimulation of oxygen consumption by high glucose was found to be independent of extracellular Ca^{2+}

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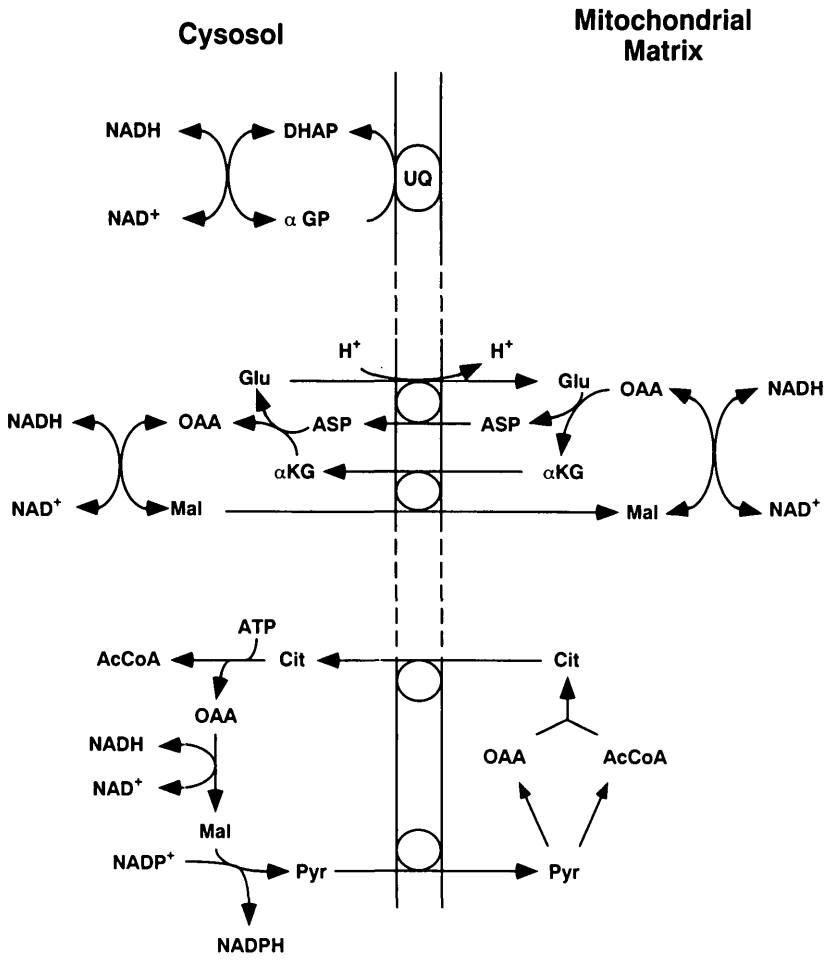


FIG. 10. Hydrogen and acetyl shuttles across the mitochondrial inner membrane. The glycero-P shuttle and the malate-aspartate shuttles serve to oxidize cytosolic NADH, and the citrate-pyruvate shuttle allows the transfer of acetyl units from the mitochondrial matrix to the cytosol. The function of the former two shuttles readily explains the glucose-induced respiratory burst, and the latter provides acetyl units for glucose-induced malonyl-CoA synthesis and the switch to glucose as exclusive fuel.

and of isobutylmethylxanthine (IBMX), which was also included in the prototype experiment depicted here in order to augment the secretory response (G. Li, F.M.M., unpublished observations). Polarographic studies with suspensions of cultured β-HC-9 cells show that the glucose concentration-dependency curve of the respiratory burst has characteristics similar to those of glycolysis and glucose oxidation, most importantly a high K_m of ~9.0 mmol/l (Fig. 9) (Y. Liang, L.-Q. Wang, C. Buettger, N. Doliba, F.M.M., unpublished observations). Crucial components of the high-K_m system of glucose metabolism of β-HC-9 cells are inducible by high glucose, as found to be true for normal β-cells. The glucose induction of the respiratory burst is paralleled by similar inductions of high-K_m glucose phosphorylation, glucose use, glucose oxidation, and insulin release. Remarkably, there is no effect of high glucose on hexose transport. It is very instructive that β-TC-3 cells lack glucose inducibility of the respiratory burst stimulated by glucose. In these cells, GK regulation of glycolysis has been lost because of the predominant expression of hexokinase and lactate dehydrogenase. These results strengthen our view that β-HC lines may be used as ideal models for the study of glucose metabolism in β-cells and in particular of the GK glucose sensor paradigm.

The glucose-stimulated respiratory burst is apparently independent of extracellular Ca^{2+} (30,36). This phenomenon is of crucial importance for an understanding of β-cell glucose metabolism. We hypothesize that glycolysis converts each glucose molecule to two pyruvate molecules with the concomitant generation of two NADH and two ATP. Based on the reports of MacDonald (21) and Dukes et al. (49), we further assume that NADH is nearly quantitatively shuttled into mitochondria. A large fraction of pyruvate is thought to leave the cell as such (rather than as lactate) or, after transamination to alanine, might be carboxylated to oxaloacetate or malate and oxidatively decarboxylated to acetyl-CoA. It is likely, however, that transamination and efflux predominate. These assumptions are made because β-cells lack lactate dehydrogenase and because pyruvate does not stimulate insulin release. Most of the oxidation of pyruvate to CO_2 that can be readily demonstrated with intact islets could be explained by non-β-cells. NADH is shuttled into the respiratory chain by the α-glycero-P and the malate/aspartate shuttles (Fig. 10). The relative importance of these two shuttles has resisted quantitation in β-cells. It seems, however, that the α-glycero-P shuttle is more favorable thermodynamically (50). The malate/aspartate shuttle is readily reversible, but hydrogen transport into the mitochondrial matrix is probably facilitated because its component glutamate/aspartate transporter is electrophoretic (50). The remarkably high activity of the glycero-P oxidase in β-cells, discovered by MacDonald (21), and the malate/aspartate shuttle may serve to prime the oxidative system with glycolytic NADH. In fact, the respiratory burst following high glucose could be fully explained if glycolytic NADH were quantitatively oxidized. Balance studies with β-HC cells suggest that this is close to the truth, because fractional glucose oxidation is very small and because only 15% of the metabolized glucose is found as lactate (Table 1). Oxidation of glycolytic NADH and of a small fraction of pyruvate fully explains the steady-state oxygen consumption in the pres-

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TABLE 1
Glucose flux and ATP generation in β-HC-9 and β-TC-3 cells

		Modeled data					
					ATP yield		
	Actual data	Glycolytic NADH yield	Specific O_2 consumption	Lactate formation	Glycolytic	Mitochondrial	Total
β-HC cells							
Glucose usage	1.83	3.66	1.66	—	3.66	8.30	—
Lactate production	0.61	—	—	0.38	—	—	18.71
Glucose oxidation	0.27	—	1.35	—	—	6.75	—
O_2 consumption	3.01	—	—	—	—	—	—
β-TC cells							
Glucose usage	5.33	10.66	1.77	—	10.66	8.85	—
Lactate production	7.09	—	—	7.12	—	—	—
Glucose oxidation	0.26	—	1.30	—	—	6.50	26.02
O_2 consumption	3.07	—	—	—	—	—	—

Data are given in mmol or l cell H_2O/min. The actual values of glucose usage, glucose oxidation, and glucose-induced O_2 consumption were determined at 20 mmol/l glucose. To calculate specific O_2 consumption, it was assumed that glucose was the exclusive fuel. To calculate lactate formation, it was assumed that respiration limits the oxidation of cytosolic NADH.

ence of 20 mmol/l glucose. To explain the results of the balance studies it is also necessary to assume that a complete switch is made from endogenous fuel (most likely fatty acids and glutamate) to glycolytic NADH and pyruvate (see below for mechanisms). It is therefore false to state that glucose stimulation of insulin secretion is independent of the citric acid cycle (49). First, the cycle needs to operate to provide NADH and $FADH_2$ for fueling basal respiration, and the rate of the entire cycle or of certain cycle segments may even increase in the presence of high glucose. Furthermore, the F_oF_1 ATPase, or ATP synthase, driven by ΔpH and $\Delta\psi$ across the mitochondrial membrane, does not distinguish between protons from different substrates (28). The citric acid cycle therefore plays a central role for stimulation of insulin secretion by all fuel molecules. It probably operates synergistically with glycolysis and might be enhanced after membrane depolarization, enhanced Ca^{2+} influx, activation of the three Ca^{2+}-activated mitochondrial dehydrogenases, and provision of ample substrate. It is very likely that increased availability of pyruvate, its carboxylation to oxaloacetate or malate, and provision of α-ketoglutarate as product of transamination (glutamate → pyruvate) contribute to supply substrate for the citric acid cycle during glucose stimulation of β-cells. The enhancement of pyruvate oxidation is probably associated and facilitated by switching off fatty acid oxidation via the well-known malonyl-CoA regulation of palmityl-CoA carnitine transferase (Fig. 1*B*). Malonyl-CoA levels of islet tissue and clonal β-cells are known to rise after glucose exposure (40–42). The surge of malonyl-CoA may be the result of enhanced pyruvate carboxylase activity, increased citrate synthesis and efflux from the mitochondria, and activation of citrate lyase in the cytosol (Figs. 1*B* and 10). Many of these processes remain to be fully explored in β-cells. It needs to be realized, however, that β-cells lack fatty acid synthesis (51), that malonyl-CoA is not a precursor for lipid synthesis, and that its turnover may be very slow (see below). Removal of extracellular Ca^{2+} inhibits glucose oxidation but has little influence on the glucose-induced respiratory burst (36,52). It is possible that Ca^{2+} lack curbs the glucose-induced metabolic switch from fatty acids as fuel for the cycle to the alternate fuel pyruvate without preventing the respiratory burst. Much needs to be learned about these subtle regulatory processes to fully understand the multiple effects of glucose on β-cell oxidative metabolism. Results of measurements of oxygen consumption of β-HC-9 and β-TC-3 cells in suspension or in a perifusion system as described above allow us to estimate the rate of ATP generation as a function of glucose concentration and to relate this parameter to insulin release (Fig. 11). The rate of ATP production has to reach a threshold before insulin release is initiated. We hypothesize that this threshold is determined by the relationship between energy supply and demand. As supply begins to predominate, the energy potential, or ΔG of ATP hydrolysis, increases sharply, providing the signal for insulin release. It needs to be realized that ATP production and use nevertheless remain balanced; merely the energy potential changes as glucose levels increase. It must be assumed, as stated, that the increase of ATP production is matched by glucose stimulation of ATP use (52). One can make only qualitative statements about the reactions that consume ATP. Increased biosynthetic activity, insulin secretion, and ion pumping are probably responsible for increased ATP use. For example, it may not be coincidence that the glucose-induced respiratory burst and the alkalinization of the cell are both blocked when HCO_3^- is eliminated from the media (30).

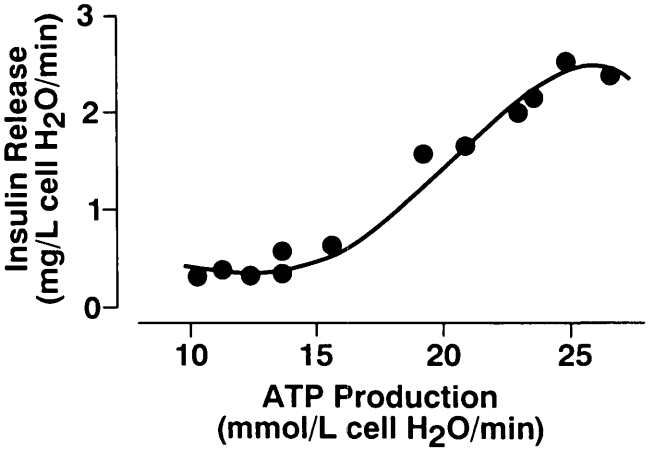
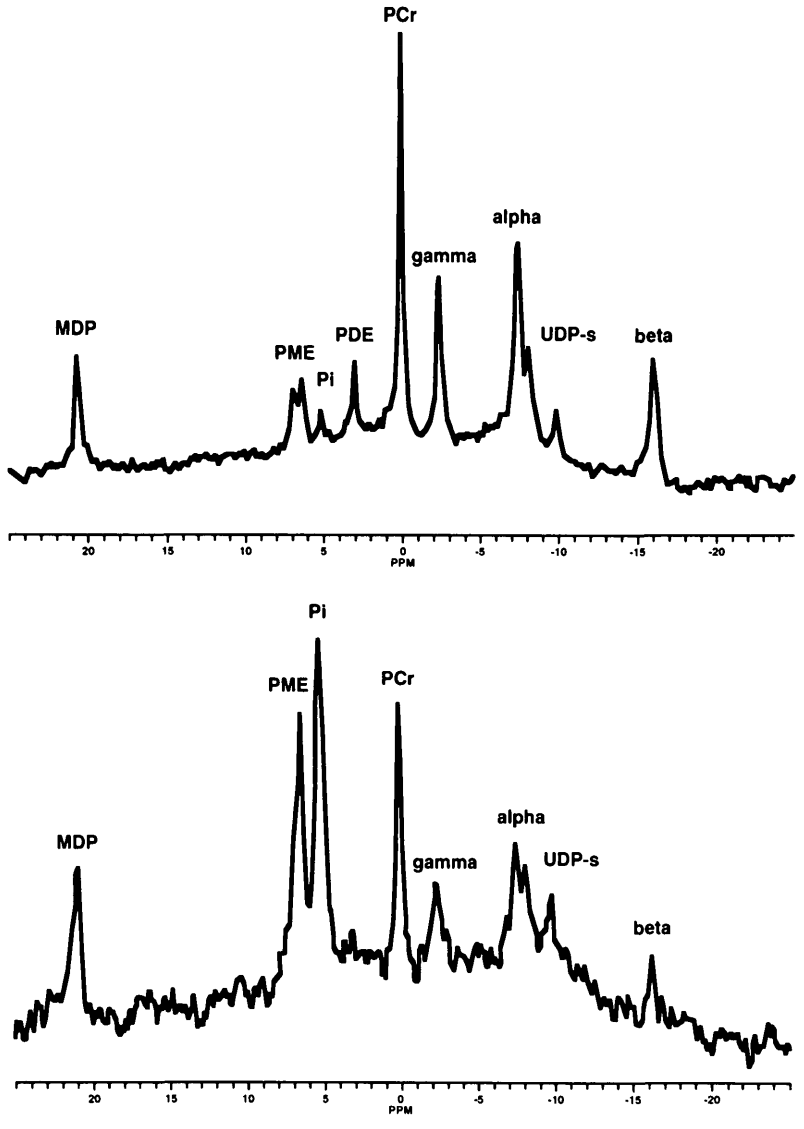


FIG. 11. ATP production and insulin release as a function of glucose concentration. ATP production was calculated from glucose usage, lactate formation, glucose oxidation, and O_2 consumption (Table 1) and was related to measured insulin release (Fig. 9). The ATP threshold for insulin release is ~16 mmol or l cell H_2O/min.

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^{31}P NMR spectra of β-TC3 (top) and β-HC9 (bottom) cells

FIG. 12. ^{31}NMR spectra of β-HC-9 and β-HT-3 cells. Large batches of creatine-loaded β-cells (~0.5 billion cells) were incorporated in micro agarose beads and perifused with low and high glucose at 37°C while observed in the magnet. Typical control spectra are shown. MDP, internal standard; PME, phosphomonoester; PDE, phosphodiester; PCr, phosphocreatine; alpha, beta, gamma, α-, β-, γ-phosphorus of ATP; UDPS, UDP sugars.

ADENINE NUCLEOTIDES AND LIPID-RELATED METABOLITES AS METABOLIC COUPLING FACTORS

High glucose increases the ATP mass-action ratio (ATP/ADP × P_i) as an expression of increased energy potential (40,52). The classic paper by Cook and Hales (53) proposed that ATP is the actual metabolic coupling factor because of its ability to inhibit K^+ channels and thus depolarize cells. However, ATP is not a plausible coupling factor because its levels change little when islets or β-cells are exposed to physiological alterations of glucose levels (34). It has been suggested that ATP levels in a subcompartment close to the cell membrane may change but that this change might escape detection. It has also been speculated that ATP changes are masked by the large stable pool of ATP that is sequestered in secretory granules or that they are obscured by oscillatory metabolic patterns. This may be true. However, an equally plausible alternate hypothesis proposes that free ADP rather than ATP serves as coupling factor, because Mg-ADP has been shown to activate the K^+ channel such that a lowering of the ADP level due to enhanced metabolism may result in inhibition of the channel (33). Free ADP levels can be calculated through the creatine kinase equilibrium (34,54,55) and pH from the chemical shift of the P_i peak in reference to the P-creatine peak. Using large batches of perifused β-TC-3 or β-HC-9 cells and ^{31}P–nuclear magnetic resonance (NMR) technology, it was possible to continuously monitor phosphorus-containing compounds (Fig. 12) and use the data to calculate cytosolic pH and free ADP levels under control conditions and after stimulation of insulin release with high glucose (54,55). Under baseline conditions, free ADP levels were ~35 μmol/l and fell to about 20 μmol/l after a switch to high glucose (Fig. 13). Such a change could explain the inhibition of the K^+ channel, considering that the K_a of Mg-ADP for the channel is about 50 μmol/l. It is remarkable that ATP levels did not change in this system when perfusate glucose was increased. The mass-action ratio of ATP and the free energy of hydrolysis of ATP are, however, increased significantly. It is interesting and puzzling that a mixture of 20 amino acids equivalent to 25 mmol/l glucose in stimulating insulin release in the presence of low glucose and 0.1 mmol/l IBMX did not cause a fall of free ADP. β-TC-3 and β-HC-9 cells behaved similarly in this regard. The biochemical basis of this difference and the metabolic coupling factor or factors that operate when amino acids instead of glucose serve as stimulus remain elusive.

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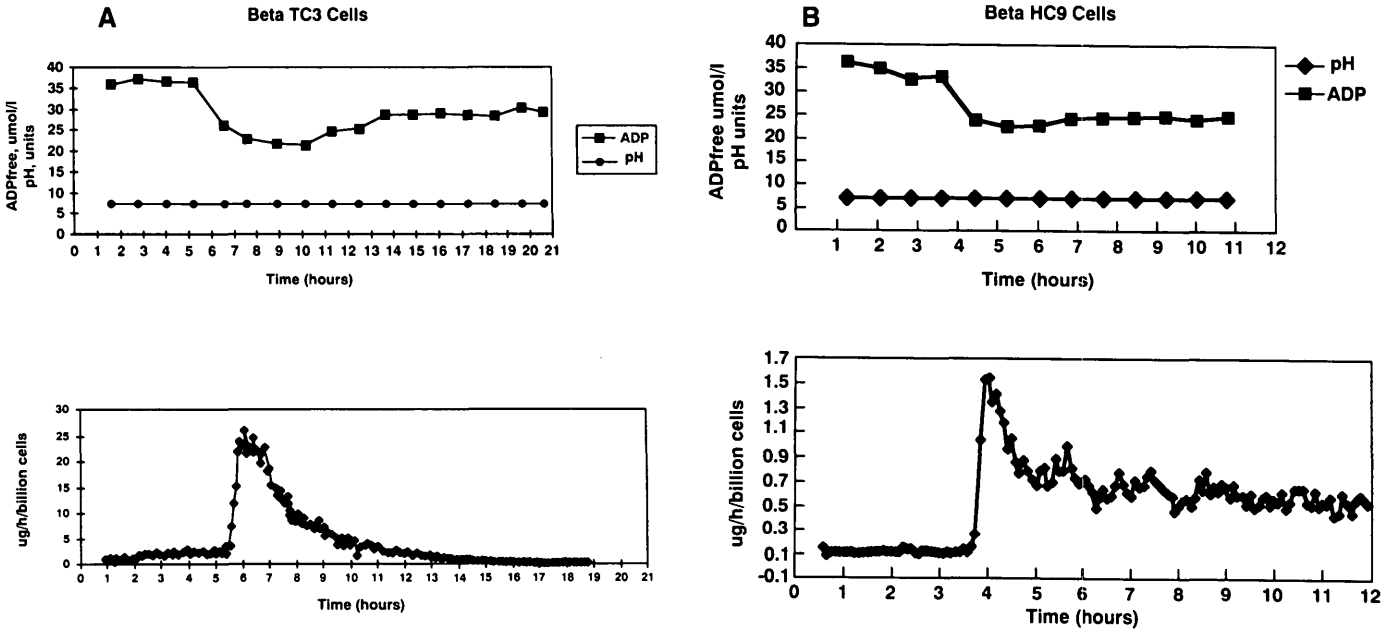


FIG. 13. Free ADP and pH in perifused β-TC-3 and β-HC-9 cells stimulated with glucose to release insulin. β-TC-3 cells were perifused with 0.2 mmol/l glucose containing 0.1 mmol/l IBMX in the control period and were stimulated by raising the glucose to 25 mmol/l (*A*). β-HC-9 cells were perifused with 3 mmol/l glucose and 0.1 mmol/l IBMX in the control period and were then stimulated by raising glucose to 25 mmol/l (*B*). Typical results are shown.

These data are consistent with a model that assigns to free Mg-ADP the role of metabolic coupling factor when glucose is the stimulus. There are at least four targets of an alteration of the ATP mass-action ratio or of Mg-ADP: *1*) the adenine nucleotide–sensitive K channel (33,53); *2*) the calcium ATPase of the endoplasmic reticulum (56); *3*) ATP-sensitive and Ca^{2+}-independent phospholipase, which generates arachidonic acid as a second messenger (57); and *4*) the process of exocytosis itself, which appears to be enhanced by a high ATP mass-action ratio (58). Any one or a combination of the first three effects could initiate the Ca^{2+} changes that are thought to initiate insulin release. More importantly, however, all four processes could be activated in a manner that does not require cytosolic Ca^{2+} levels to change and could thus initiate the hormone release process. There is a considerable body of evidence that intermediates of lipid metabolism may serve as coupling factors that connect glucose metabolism to insulin secretion (39–43) in a manner that may function in parallel or may be an obligatory direct link of the chain connecting glucose with insulin release. A glucose-induced elevation of malonyl-CoA is central to this process and has been demonstrated in HIT cells (40,41) and in isolated perifused pancreatic islets (42). The temporal profile of the malonyl-CoA after stimulation with glucose or a potent fuel mixture of glucose, glutamine, pyruvate, and lactate was biphasic with a pronounced first peak (Fig. 14). However, acyl-CoA levels rose more slowly with time. The results are consistent with a role of these intermediates as metabolic coupling factors. As discussed earlier, the rise of malonyl-CoA may serve to curb fatty acid oxidation and to facilitate glucose oxidation. One consequence of this switch might be the accumulation of cytosolic acyl-CoA and of DAG, which are thought to activate protein kinase C. This enzyme seems to participate in exocytosis of insulin (Fig. 1*B*). Hydroxycitrate, an inhibitor of citrate lyase, was used to further test this concept (43). This antimetabolite inhibited glucose-induced insulin release, and the inhibition was overcome by palmitate, results consistent with the hypothesis. It remains to be seen whether hydroxycitrate is indeed devoid of side effects as assumed (43). Intracellular signaling by acyl-CoA and DAG in the course of glucose-stimulated insulin release remains highly speculative, contrasting with the more firmly established role of adenine nucleotides. Whatever its importance, it is emphasized in this review that the glucose sensor GK determines the levels of critical signaling molecules related to lipid metabolism as effectively as it governs the ATP mass-action ratio. Both signaling pathways are controlled by one and the same glucose sensor.

It must be appreciated at this point of the discussion that the second messenger Ca^{2+} may amplify the glucose-derived metabolic signal by activating or enhancing phospholipases, adenylate cyclase, and protein kinases, which appear to be significant participants in insulin secretion and biosynthesis (31,37,39,40).

GK: MOLECULAR ASPECTS

The preceding paragraphs have shown that the β-cell glucostat derives its physiological sensitive range of 4 to 10 mmol/l glucose for its glucose-dependent functions from the GK molecule. A brief account of the molecular characteristics of this enzyme is therefore in order. A scheme of the GK gene of humans provides fundamental information about the structure of the enzyme and its isoforms (59) (Fig. 15). A dual promoter arrangement with upstream islet-specific exon 1-B and downstream liver-specific exon 1-L was found in the genes of rats, mice, and humans alike. The mature GK mRNAs are assembled from the tissue-specific exons 2–10 in a manner illustrated in Fig. 15. The amino acid sequences of islet and liver GK of rat and human have been deduced from the sequences of the respective cDNAs. The sequence of islet GK has 465 amino acid residues and a calculated molecular weight of ~52,000 Da. The sequences are highly conserved

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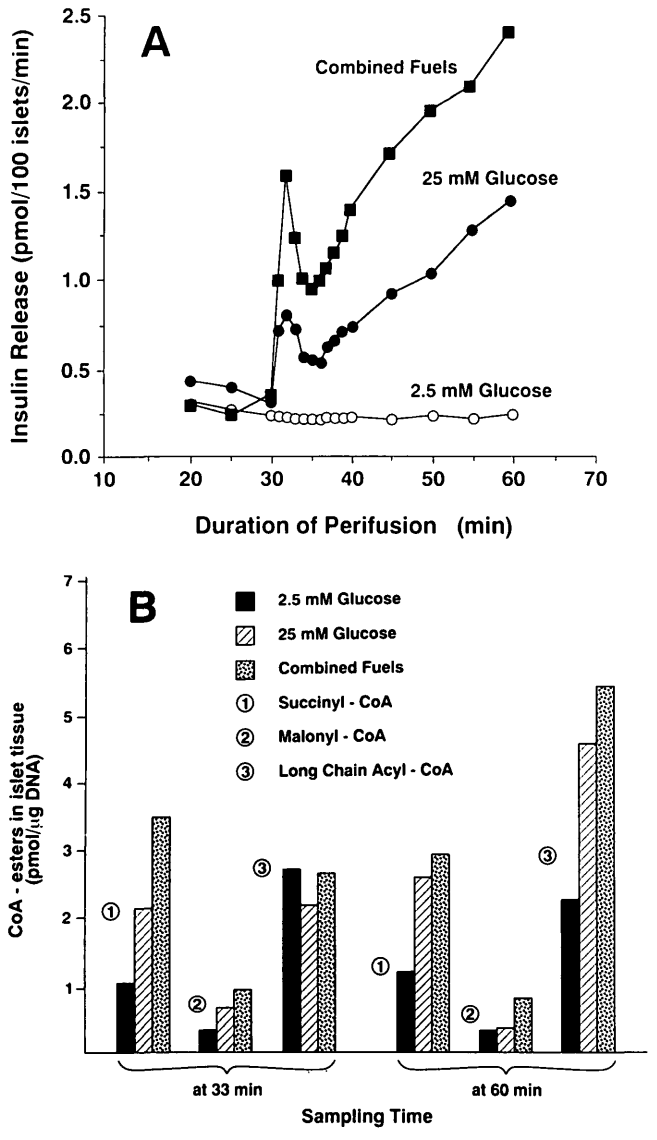


FIG. 14. High glucose increases malonyl-CoA levels in isolated perifused pancreatic islet tissue. Insulin release is related to the results of intermediate measurements at two strategic sampling times. Data from Liang and Matschinsky (42).

with 98% identity for human and rat liver isoforms. Liver and islet β-cell forms of the enzyme differ by a short highly charged NH_2-terminal peptide stretch of 14–16 amino acids encoded by the β-cell–specific leader exon. A specific function of this sequence, if any, remains unknown. Since mammalian GK exhibits a high degree of identity and conservative amino acid substitution compared with yeast hexokinases, the known X-ray structure of yeast hexokinase B has been used to model mammalian GK and to project the locations of functionally critical amino acid residues and their interactions with the hydroxyls of the glucose molecule (19): Asn^{204} (−OH3 and 4), Asp^{205} (−OH4), Gly^{229} (−OH6), Glu^{290} (−OH1), Glu^{256} (−OH2 and 3), and Asn^{231} (−OH4). These six amino acids are thought to be involved in creating the glucose binding site of human GK. The location and configuration of the ATP binding site has not been established. Long-chain acyl-CoA binds to an allosteric site, which appears to overlap with the binding site for the 62-kDa fructose-6-P–dependent regulatory protein, as suggested by kinetic evidence (60). GK regulatory protein increases the GK K_m for glucose in the presence of fructose-6-P with little

TABLE 2
Substrates and inhibitors of recombinant human islet glucokinase GST fusion protein

Sugar	Activity (U/mg protein)	K_m (mmol/l)	n_H	K_i (mmol/l)
D-glucose	134	5.3	1.41	N/A
D-mannose	175	24	1.38	10
D-fructose	260	424	1.31	NM
D-2-deoxyglucose	70	85	1.30	10
D-glucosamine	6.9	6.6	1.04	1.7
D-mannosamine	0.26	16	0.82	3.8
D-mannoheptulose	0.24	7.5	1.01	0.78
D-*N*-acetylglucosamine	ND	ND	ND	0.14
Palmityl-CoA	N/A	N/A	N/A	7.5×10^{-3}
Stearyl-CoA	N/A	N/A	N/A	2.6×10^{-3}
Oleyl-CoA	N/A	N/A	N/A	8.6×10^{-3}

All kinetic constants were obtained spectrophotometrically at 30°C. A Q_{10} of 2 was used to calculate the activity for 37°C. The K_i was determined with racemic D-glucose and Mg-ATP as substrate. Glucosamine phosphorylation is strongly and noncompetitively inhibited at high glucosamine concentration and reaches zero at ~750 mmol/l. 1 U = 1 μmol product/min. N/A, not applicable; ND, not detectable; NM, not measured.

change of V_{max}. The effects of regulatory protein and palmityl-CoA are competitive. It is thought that binding of regulatory protein and/or acyl-CoA to an allosteric site prevents the glucose-induced conformational change that is believed to be essential for enzymatic catalysis. The exploration of GK kinetics and structure has been facilitated by the ample availability of recombinant enzyme. A rapid single-step chromatographic purification scheme using the glutathione S-transferase (GST)–GK fusion protein has greatly facilitated the research (20). Virtually pure enzyme is obtained and the NH_2-terminal GST tail has no known effects on enzyme function (Table 2). The concentration-dependency curve of phosphorylation of many hexoses by GK is sigmoidal, with a Hill coefficient of 0.8–1.4. Sigmoidicity was observed with glucose, mannose, fructose, and 2-DG as substrates. Glucosamine, another substrate, shows unique biphasic kinetics, which are best explained by substrate inhibition or the involvement of an allosteric inhibitor site in addition to the substrate binding site. Mannoheptulose and *N*-acetylglucosamine are potent inhibitors. Mannoheptulose may also be a weak substrate for the enzyme, but this remains to be

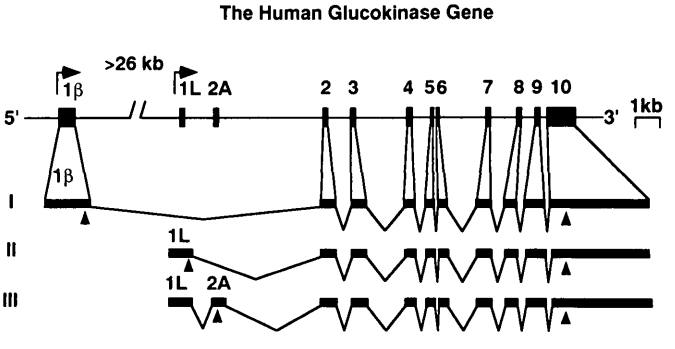


FIG. 15. The GK gene in humans. The top line shows the exon-intron organization of the genes. Filled boxes represent the exons and curved arrows the sites of transcription initiation. Lines I–III show tissue-specific initiation of transcription and alternate splicing. Arrowheads denote initiation and termination codons. I: transcription and pre-mRNA splicing in β-cells of the islets of Langerhans; II and III: transcription and pre-mRNA splicing in hepatocytes. Modified from Iynedian (59).

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TABLE 3
Glucokinase "trivia" in humans

k_{cat} 37	135/s
K_m-glucose	6 mmol/l
n_H-glucose	1.5
Pancreas GK capacity	10 nmol · pancreas^{-1} · s^{-1}
Pancreas GK protein	4 μg/pancreas
Liver GK capacity	100 μmol · liver^{-1} · s^{-1}
Liver GK protein	40 mg/liver

The values collected in this table are physical constants (items 1–3) or estimates based on reasonable assumptions.

documented by demonstration of the reaction product. Galactose is neither substrate nor inhibitor. Long-chain acyl-CoAs are potent inhibitors of human recombinant β-cell GK, suggesting that the β-cell isoform could also be inhibited by regulatory protein. Note that the three known human GK isoforms studied here as GST-fusion proteins appear to be indistinguishable kinetically. It remains to be ascertained whether a small (twofold) difference of the K_m for ATP between the human islet and the liver forms is real (20). This characteristic, if confirmed, could be significant, since the glucose K_m is a direct function of the ATP concentration in the physiological range. It is believed that GK is active as a monomer in distinction to most other allosterically regulated proteins. Three explanations have been offered for the sigmoidal rate behavior of the enzyme. It has been attributed to *1*) the formation of a random ternary complex that may deviate from classic Michaelis-Menten kinetics (61), *2*) a "mnemonic" mechanism according to which glucose binds to two forms of the free enzyme that are not in equilibrium (62), or *3*) a slow transition model (63). It remains to be seen which of these explanations, if any, applies. Whatever the biophysical basis, sigmoidicity, manifest in Hill numbers as high as 1.8, may greatly improve the sensitivity of GK to glucose in the physiological range, particularly in β-cells. Human islet GK has an apparent K_m of 5.3–7.7 mmol/l and an inflection point of 2.7 mmol/l (5,20). These values bracket precisely the physiological blood glucose range. Even small alterations of V_{max}, K_m, and n_H may have profound effects on glucose phosphorylation and glucose flux in β-cells. The GK molecule is very unstable in solution devoid of K^+, sulfhydryl reagents, and glucose, as judged by the results of activity measurements. Stabilization by glucose is concentration dependent in the physiological range of the hexose, with an ED_{50} of ~10 mmol/l. Stabilization by glucose may be extremely important biologically and in the pathogenesis of certain MODY cases, as discussed below, and could be crucial experimentally in studies of the enzyme's kinetic characteristics.

From the information given so far, several extrapolations can be made, which might be summarized as GK "trivia" for humans (Table 3). Noteworthy is the estimate of a minimal amount of ~4 μg of β-cell GK per pancreas sufficient to serve as glucose sensor in glucose homeostasis and contrasted to the 10,000-fold higher amount of the liver enzyme that serves a quantitatively significant process in the glucose economy of the organism. When blood glucose levels are in the K_m range, hepatic GK is capable of removing ~0.5 g/min of glucose from the blood stream, or clearing ~0.5 l/min of serum, whereas metabolism of 100 μg/min of glucose is enough to fuel β-cell metabolism.

Regulation of GK gene expression exists at multiple levels with differential tissue-specific emphasis (64) (Fig. 16). Use of two distinct promoters in a single gene is the basis of cell-specific regulation of GK expression. The upstream promoter governs expression in the β-cells, the pituitary, and possibly other central nervous system, autonomic nervous system, or intestinal sites, whereas the downstream promoter determines hepatic GK gene expression (65). GK gene expression in β-cells appears to be constitutive and to be regulated primarily posttranscriptionally by glucose. We hypothesize that enzyme levels are influenced directly by the glucose molecule itself, which stabilizes the enzyme. Glucose metabolism may also contribute to stabilization by influencing the redox state of essential sulfhydryls of the enzyme. GK expression in hepatocytes is, however, controlled by the insulin receptor and influenced by biotin(+), T_3(+), and glucagon(−). This tissue-specific differential regulation of GK was first demonstrated in insulinoma-bearing NEDH rats, and its significance was fully recognized (66). The insulin-producing tumor caused hyperinsulinemia and hypoglycemia in the animals. Liver GK was induced three- to fourfold, and islet GK was reduced to <20%. Removal of the tumor lowered serum insulin and reversed the hypoglycemia with an overshoot to hyperglycemia, which was accompanied by a dramatic decline of hepatic GK and a rapid rise of islet GK. The molecular basis of this differential regulation of GK gene expression in the two tissues is likely complex and remains an area of active investigation. The possibilities for regulation are numerous, as illustrated in Fig. 16. Of great interest is a 20-year-old report from the laboratory of Coleman (67), which distinguished genetically based low and high expressors of liver GK in mice. It is conceivable that constitutive expression of β-cell GK is genetically variant as well, such that low and high insulin responders to a glucose challenge might be explained on that basis (68). Because of the pivotal role of this enzyme in glucose homeostasis, it may

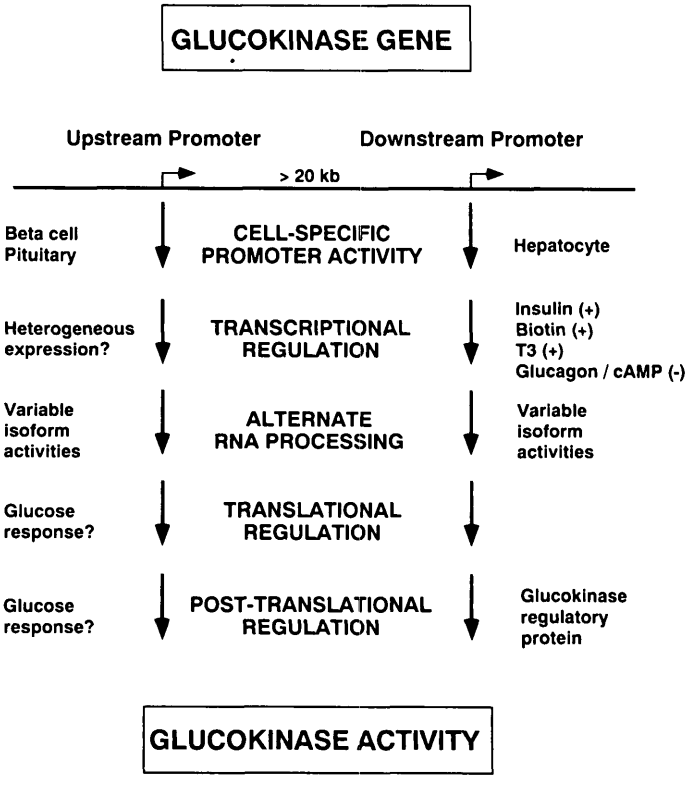


FIG. 16. Factors that determine differential tissue-specific expression of GK. From Magnuson et al. (64).

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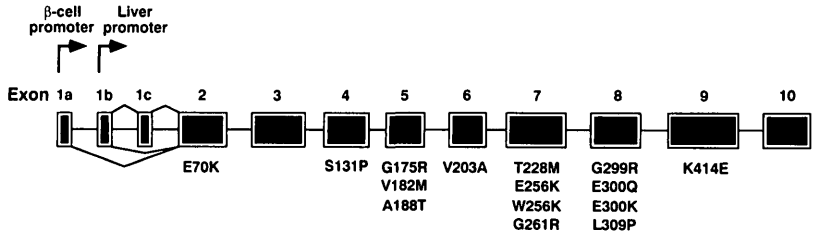


FIG. 17. Structure of the human GK gene and localization of several mutations identified in patients with MODY. Modified from Pilkis et al. (19).

prove therapeutically useful to enhance GK gene expression or the enzyme's stability pharmacologically. Approaches may have to differ for specific target tissues because of the divergent tissue-specific controls of GK gene expression.

GK MUTATIONS IN MODY AND LINKAGE ANALYSIS IN NIDDM

Because of its pivotal role in glucose homeostasis, GK stood out as an ideal candidate gene in the search for the genetic basis of NIDDM (2–5). Highly polymorphic microsatellites had been identified at three positions of the human gene, which greatly facilitated the chase for GK mutations in human pedigrees with familial diabetes using PCR-based methodology (14,15). The MODY syndrome, characterized by an early onset and by autosomal dominant transmission with high penetrance, was the most attractive condition for analysis of GK gene mutations. Indeed, close linkage of the GK locus on chromosome 7p to MODY was first described in 1992 by Forguel and colleagues (16,17) in 32 families collected in France. The genetic information resulted in a maximum logarithm of odds score between diabetes and GK of >25 for all MODY families, at a recombination fraction of 0.00. The odds in favor of linkage were thus very high ($>10^{25}$:1). Of the French families with MODY, 60% showed linkage to the GK locus. In the U.S., 1 in 2,400 individuals may be heterozygous for mutations in the GK gene (19). As many as 1% of all diabetes may therefore be explained by GK mutants, which makes the GK mutation the most common cause of NIDDM identified to date. It is also the only instance where this disease can be attributed to a clearly defined defect of glucose metabolism. Since the gene defect is transmitted dominantly, all known subjects with GK mutations appear to have one normal allele and homozygocity with total loss of GK might be lethal. It also suggests that a gene dosage mechanism could explain the phenotypes, which are expected to have tissue activities of GK that may be as low as 50% of control subjects', depending on the nature of the mutation and the tissue-specific controls of gene expression. The swift and competent work of several laboratories has identified and characterized ~30 specific mutations of the GK gene within the short period of 3 years since the discovery of linkage (19) (Fig. 17). A quick survey showed that the mutations cause alterations of V_{max}, K_m values for glucose and ATP, n_H, and enzyme stability (5,17,19,20) (Table 4). Because the GK defect involves at least two tissues participating in glucose homeostasis, it is currently not clear what the relative contributions of liver and pancreas to the clinical picture might be. In the worst-case scenario, GK activity in β-cells is expected to be cut to half of normal, and because of the concomitant decrease of insulin release, it is projected that hepatic GK is also reduced. But it is difficult to predict by how much because regulation of hepatic GK gene expression appears to be more complex than that of β-cells. It is not unlikely that mutations will be found in the future that lead to a specific lesion of either the liver or the β-cells. Work with transgenic animals promises to illuminate the complex pathophysiology and biochemistry of GK gene mutations in MODY patients (69,70). Targeted disruption of GK gene expressions in pancreatic β-cells of mice did indeed cause diabetes (70). Clinical research has shown that glucose-induced insulin release is reduced in all MODY patients with GK mutations demonstrating the β-cell enzyme deficiency (17). Since hepatic GK expression is insulin dependent, it is difficult to distinguish between an enzyme reduction due to lower gene dosage and one due to lack of insulin. Another issue arises from the lack of close parallelism of severity of disease and functional consequences of gene mutations on in vitro enzyme activity. This illustrates limitations of extrapolation from results of DNA analysis and kinetic investigations of recombinant mutant enzyme to the actual glucose phosphorylation rates in β-cells and hepatocytes of the particular patient. Much new insight into these questions emanates from thorough in vivo and in vitro studies with recombinant mutant GKs. Stability tests of the E70K and E300K GK mutants are a case in point. A diabetic phenotype had prompted the original discovery of these particular mutants. The initial survey suggested a decreased catalytic function of

TABLE 4
Kinetic characteristics of recombinant wild type and mutant human β-cell glucokinase

	V_{max} (U/mg protein)		K_m (mmol/l)		n_H	
	GK-I	GST–GK-I	GK-I	GST–GK-I	GK-I	GST–GK-I
Wild type	93	54	6	6	1.7	1.4
E70K	23	36	62	8	1.6	1.5
S131P	46	—	110	—	1.6	—
G175R	51	13	39	31	1.5	1.1
V182M	49	13	70	37	1.5	1.3
A188T	43	—	72	—	1.6	—
V203A	0.5	11	100	123	1.3	1.5
T228M	0.4	0.04,0.01	10	14.3,4.4	1.3	1.4,1.1
E256K	0.2	—	92	—	1.0	—
W257R	0.5	—	5	—	1.4	—
G261R	0.4	2.8	2.5	144	1.5	1.0
E279Q	55	—	41	—	1.5	—
G299R	0.3	—	3	—	1.6	—
E300K	33	29	25	4.7	1.4	1.3
E300Q	100	—	20	—	1.5	—
L309P	0.9	1.0,0.5	22	24,32	1.6	1.4,1.3
K414E	94	8.4,7.7	80	29,17	1.5	1.2,1.4

Mean values are recorded for most cases, except for n <3, in which case the individual values are given. Activities were measured at 30°C. Activities are comparable for GK-I and GST–GK-I; the apparent difference is explained by the higher molecular weight of the fusion protein. GK-I data from Pilkis et al. (19), GST–GK-I data from Liang et al. (20).

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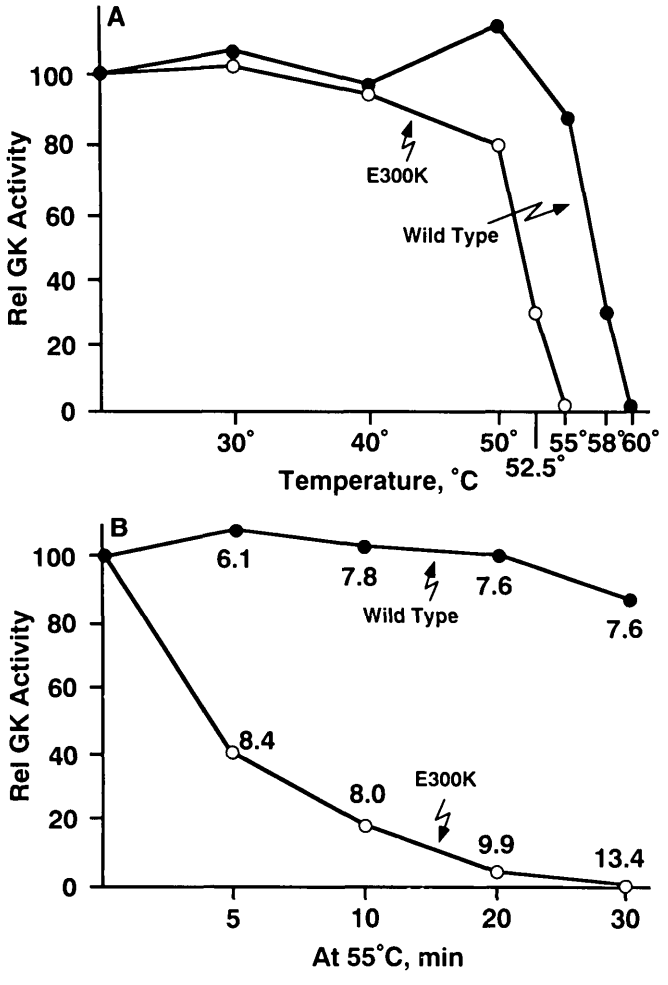


FIG. 18. Increased temperature lability of GST-GK mutant E300K. *A:* the enzymes were incubated for 30 min before the assay at different temperatures in the presence of 50 mmol/l Tris-HCl (pH 8.0), 10 mmol/l glutathione, 50 mmol/l glucose, 5 mmol/l dithiothreitol, 30% glycerol, and 200 mmol/l KCl. The temperatures are indicated. *B:* the enzymes were incubated at 55°C for different lengths of time before the assay in the storage buffer described above. The activity of the wild type was 69 U/mg, and its K_m measured 6 mmol/l. The activity of the mutant was 62 U/mg, and its K_m measured 7 mmol/l. Relative activities are given as well as the K_m values for the time-course study. The data depict the results of recent repeats of experiments described in Liang et al. (20). Note the difference of the present kinetic data to those given in Table 3, which show reduced V_{max} values. Increased speed of enzyme preparation and improved assay conditions are the likely explanation for the difference. Comparable observations were made with preparations of mutant E70K. An example of three experiments with comparable outcomes is shown. The apparent abnormality of the kinetic constants as given in Table 4 is probably due to denaturation during purification and storage of a typical preparation.

these mutants (Table 4) (19,20). A more careful analysis showed, however, that the kinetic constants of these two mutants were normal in optimal enzyme preparations, yet the enzymes were aberrant in standard stability tests (Fig. 18). E70K and E300K were inactivated much faster than the wild type. The lower stability of these mutants is particularly striking at 37°C in the presence of 15 mmol/l glucose. Extrapolations to the in vivo situation are tempting, but additional studies are needed to test whether K70K and E300K GK mutants are indeed less stable than the wild type in the intact β-cell. It is also obvious that direct analysis of GK activities in hepatocytes and islet tissue of MODY patients could help answer many quantitative issues. A highly specific, sensitive, and quantitative oil-well method is available that would allow the determination of the enzyme in minute biopsy specimens of <100 cells (46). In summary, it appears justified to distinguish between low-activity and low-stability mutants of the GK gene in MODY, a mechanistically and pathogenetically significant distinction. It is tentatively proposed that mutants E70K, E300K, and (by extrapolation) E300Q represent low-stability mutants, whereas V203A and G261R appear to be clear examples of low-activity mutants. The distinction may become blurred in certain cases because the substrate glucose, usually included in the storage buffer of recombinant enzyme, is an effective stabilizer. Extrapolating to the intact β-cell glucose may also be a less effective stabilizer in situ in cases where mutations increase the GK K_m for glucose. The results of published reports (19,20) need to be reassessed in view of these new findings and considerations.

Concannon and colleagues (71) have detected linkage and association to the GK locus in IDDM using family and population screening (71). The enzyme's structure was apparently not altered in afflicted subjects. This result may seem surprising because the disease is thought to arise from the destruction of β-cells due to activation of the autoimmune system. There are several plausible explanations for the apparent GK connection to IDDM. First, linkage and association obviously do not prove direct GK involvement. Second, differences in constitutive expression of GK could be involved and might compromise β-cells that constitutively express low levels of the enzyme, rendering them more susceptible to immune attack (67 and above). Third, GK may play a role in glucose metabolism in a critical set of lymphocytes or macrophages that participate in the immune destruction of the β-cells. Currently, there is no information to distinguish between these three possibilities or to postulate other mechanisms. The report is, however, another indication of the central position of GK in β-cell biology.

F_OF_1 ATPase DEFECT AS CAUSE OF DIABETES IN BHE_{CDB} RATS

The search for candidate genes for diabetes should continue to take its cues from the physiological chemistry of pancreatic β-cells, particularly from the knowledge about regulatory steps of glucose and energy metabolism. The identification of GK as diabetes gene is one case where this approach has paid off handsomely. Mutations of the insulin receptor and insulin genes were identified much earlier by the same strategy, and their pathogenetic significance is quantitatively similar to that of the GK gene mutations in the explanation of NIDDM, which all together clarify probably <5% of NIDDM cases. The search for diabetes genes must therefore continue vigorously. The random gene approach using a technique called positional cloning is taken by the gene industry (72). It has high predictability of ultimate success but is associated with exorbitant cost and much drudgery. The other is the candidate gene approach pursued by physiologists and clinicians, with GK as case in point (16–18). The deliberations of the preceding paragraphs highlight several additional β-cell enzymes of glucose, lipid, and energy metabolism that are plausible candidates. Of these possibilities, we will discuss here the mitochondrial ATP synthase, or F_OF_1 ATPase, because of its indispensable collective role and high control strength in the transduction of fuel stimulation of insulin release (28,29,73). ATP synthase serves as the exit gatekeeper of energy metabolism in analogy to GK as the entrance gatekeeper of glucose metab-

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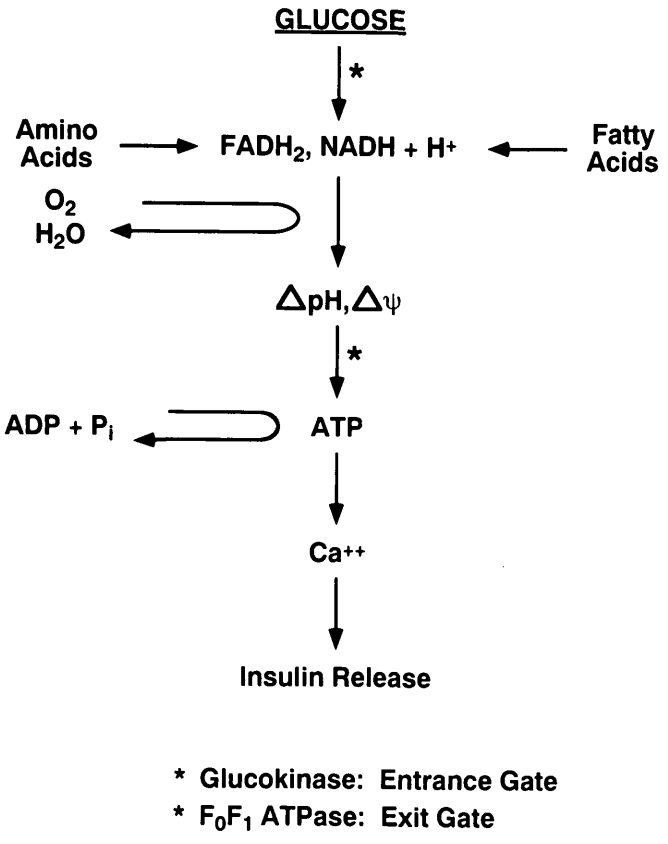


FIG. 19. GK and ATP synthase as critical steps in glucose stimulation of insulin release.

olism (Fig. 19). ATP production is a function of the F_oF_1 ATPase activity, the ΔpH and Δψ across the mitochondrial membrane, availability of acceptor ADP, and adenine nucleotide translocator activity. We were therefore on the lookout for a suitable model system to assess the involvement of this essential step. Berdanier (74) had postulated that NIDDM in the BHE_{cdb} rat may be due to a defect of ATP production of liver mitochondria (74). In collaboration with Dr. Berdanier, we were able to show that BHE_{cdb} rats have an islet defect as well, and we reason here that this islet impairment might be explained by reduced ATP production, because we already observed that glucose transport of islets was normal and that islet GK was actually enhanced in these animals (75). Our interest was much excited by the identification by Berdanier and colleagues (76) of a mutation in the F_o component of the enzyme encoded by mitochondrial DNA. A change of the aspartate codon (GAC) to the asparagine codon (AAC) in subunit 6 of the BHE_{cdb} F_o ATPase gene may alter the critical polar pocket of the enzyme and decrease efficiency of ATP synthesis. The results are greatly strengthened by a previous report from an independent group of an indistinguishable mutation at the same position of the F_o ATPase in a diabetic rat colony maintained by WIGA, Sulzfed, Germany (77,78). We hypothesized that β-cells might share the primary mitochondrial defect. Islets from young BHE_{cdb} rats might therefore show abnormal functional behavior. In particular, they might adapt insufficiently to a protracted overload with high glucose in organ culture in contrast to normal islets. This prediction was tested using 75- to 105-day-old BHE_{cdb} rats of either sex and respective Sprague-Dawley controls as islet donors. BHE_{cdb} rats are only marginally diabetic at this age, if at all. Isolated BHE_{cdb} islets were microscopically indistinguishable from control islets and had the same DNA contents. We compared insulin release due to glucose or amino acids by freshly isolated islets and by islets cultured for 7 days at 5 or 10 mmol/l glucose using a perifusion setup that allowed stimulation by a glucose or amino acid ramp such that the dose-response curve over a wide range (0–50 mmol/l) could be obtained in a single experiment. Our findings are consistent with such a primary abnormality of β-cells (Fig. 20).

Freshly isolated islets from BHE_{cdb} rats hypersecreted insulin compared with islets from Sprague-Dawley controls. This result may seem surprising, but it is not inconsistent with many observations about the ontogeny of NIDDM showing early hyperinsulinemia. However, the adaptive response to 7-day culture in 10 mmol/l glucose was reduced in islets from BHE_{cdb} rats, and the dose-response curve for glucose stimulation showed a shift to the left such that the threshold was lowered to ~2.5 mmol/l. This contrasted with the results from islets of normal rats, which adapted to high glucose in an expected manner. In particular, the dose-response relationship, with a threshold between 5 and 7 mmol/l, was precisely preserved and secretory capacity was

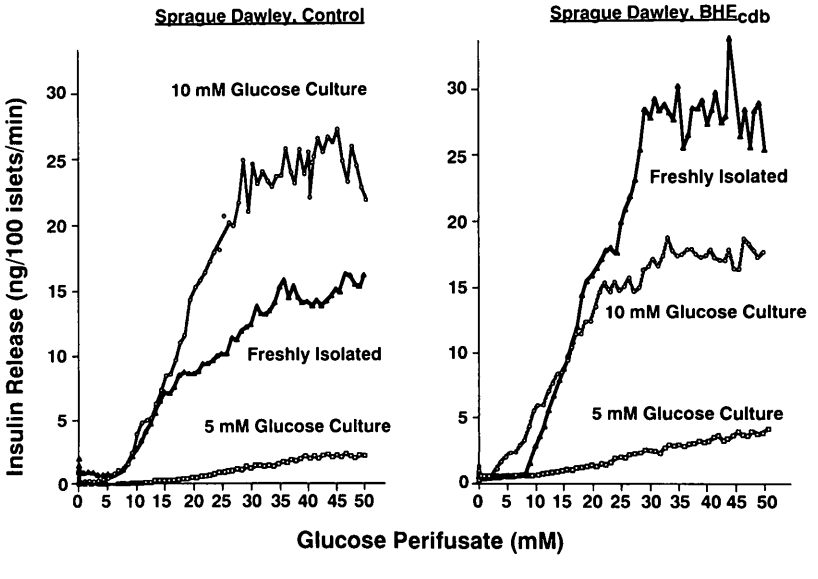


FIG. 20. Maladaption of islets from young BHE_{cdb} rats to culture at 10 mmol/l glucose. Islets from 75- to 100-day-old BHE_{cdb} rats and from suitable controls were perifused either immediately after isolation or after culturing for 7 days at low (5 mmol/l) or high (10 mmol/l) glucose. Glucose stimulation was performed with a ramp function with a change of glucose from 0 to 50 mmol/l over a 65-min period. This test provided a glucose dose-response curve indicating the glucose threshold and the saturation levels for stimulation. Data for freshly isolated islets and cultured islets are presented. The mean of six experiments is given for each condition. The DNA contents of islets from controls and BHE_{cdb} rats were indistinguishable, virtually eliminating islet mass as an explanation for the difference in results.

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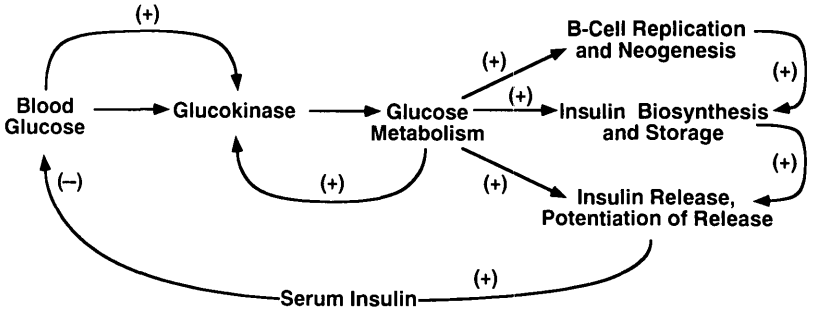


FIG. 21. Pleiotropic actions of glucose on the β-cells. Modified from Bedoya et al. (66).

greatly augmented. We thus detected alterations of an islet tissue function that seem to be the hallmark of the diabetic β-cell, that is, high baseline release and lowered maximal response when chronically exposed to high glucose. The reduced adaptive response was also manifest when islets were stimulated with an amino acid mixture, but with that stimulus, the threshold shift clearly demonstrable with glucose as stimulus was not obvious. The BHE_{cdb} β-cell maladaption to the stress of high glucose in culture might indeed be a primary phenomenon. A primary involvement of impaired ATP synthesis in β-cells is a particularly attractive explanation because there is no reason to believe that the mitochondrial genome of β-cells differs from that of liver cells. The late-onset nature of the disease manifesting itself in both the periphery and the islets may find a ready explanation from the normal age-related decline in oxidative phosphorylation function, which may be accelerated in BHE_{cdb} animals because of the mutated F_o component of the ATPase. The hypothesis is plausible and eminently testable and is of great potential relevance for an increased understanding of NIDDM in humans. Recent reports of mitochondrial in humans underscore the relevance of the BHE_{cdb} and WIGA models of diabetes (79,80).

PERSPECTIVE

The goal of this presentation was to sketch only the most distinguishing features of the β-cell's metabolic pathways from blood glucose to ATP as the common product of energy metabolism and to other metabolic signals (e.g., malonyl-CoA). The purpose was also to describe the effect that an altered ATP mass-action ratio has on critical membrane ion channels and pumps and to delineate the effects that malonyl-CoA may have on intracellular signal transduction. These discussions avoided on purpose the baffling hierarchal complexity of layer upon layer of many other controls that appear to impinge on the β-cell (31,37,39,40,81). These include metabolic oscillations, ion channels and pumps (e.g., K^+-ATP channel and the L-type Ca^{2+} channel), autocrine factors (e.g., ATP and eicosanoids), hormones [e.g., GLP-I(7–37) and somatostatin], neurotransmitters (e.g., acetylcholine and norepinephrine), exocytosis, cellular diversity, cell-cell interactions through gap junctions, etc. Whatever the role of these and countless other modifying systems, they must be understood within the framework of β-cell glucose metabolism as discussed here. Each one of them would, however, deserve a dedicated chapter to be fully appreciated. It is important to appreciate that enhancement of metabolism by glucose not only stimulates insulin secretion and insulin biosynthesis, but also appears to initiate the complex process of β-cell adaptation to fuel oversupply, which includes potentiation, hypertrophy, hyperplasia, and neogenesis of β-cells (66,82,83) (Fig. 21). A defect of glucose sensing due to a GK defect or impaired metabolic coupling due to a mitochondrial F_oF_1 ATPase mutation would prevent these crucial physiological adaptive responses and hasten the development of a diabetic phenotype. It is truly amazing that this complex molecular machine of the β-cell functions so well, such that only a few percent of the human population are diseased because of its malfunction. But it is not surprising that the adaptability of the β-cells to overstimulation due to excessive fuel loads is limited, as is manifest in the dramatic increase of its demise in an affluent, overfed, and inactive society with an increasing incidence of diabetes. Recent related results from animal models are striking and offer plausible mechanistic explanations (84–86). The incidence of diabetes might be greatly augmented by a condition that might be called an "affluenza" of the pancreatic β-cell. Using the two examples of the GK and F_oF_1 ATPase genes illustrates how a critical mutation in glucose and energy metabolism might impair β-cell function and may lead to frank diabetes. These genes may be well suited to study environmental factors that influence the β-cell. Taking a cue from these studies, other molecular pathogenetic mechanisms are likely to be discovered in the future. The job is formidable, since so far a molecular genetic explanation has been found for only a small percentage of <5% of NIDDM.

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